You-Ting Wu

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Education

Columbia University — New York, NY, USA
Ph.D., Applied Mathematics; M.Phil. (expected May 2027); M.S. (Feb 2026) — Sept 2024 – Aug 2028 (expected)

National Taiwan University — Taipei, Taiwan
B.Sc., Mechanical Engineering and B.Sc., Atmospheric Sciences — Sept 2018 – July 2023

Skills

Programming and HPC: Python (PyTorch, SciPy, Scikit-learn, xarray), C/C++ (numerical solver for fluid dynamics PDEs, course project), Fortran, MPI, MATLAB, Git/GitHub
Machine learning: deep learning, CNN, U-Net, LSTM, neural stochastic differential equations, forecasting, predictive modeling, out-of-sample validation
Mathematics: nonlinear dynamical systems, numerical analysis, linear algebra, numerical PDEs
Statistics: regression, hypothesis testing and significance analysis, sampling and bootstrap methods, variance decomposition, time series

Professional Experience

Graduate Research Assistant, Columbia University, *New York, NY, USA* — Sept 2024 – Present

- Developing CNN and U-Net networks in Python that separate forced responses (greenhouse gas, aerosol, natural) from internal variability in large climate-model ensembles, tested on held-out models to assess generalization under distribution shift
- Building a neural stochastic differential equation (neural SDE) emulator of internal variability to estimate how rare the observed 2007–2026 Arctic sea-ice pause is; next: calibrated uncertainty (conformal prediction) and robust training across models
- Built and validated CNN and LSTM models in PyTorch that forecast high-dimensional spatiotemporal indicators over 70-year with roughly 100x speedup over direct simulation; evaluated out-of-sample generalization across unseen scenarios to guard against overfitting
- Ran and modified a large Fortran/MPI codebase on university HPC clusters to generate and process a 2,000+ simulated-year, 20-member ensemble dataset for analysis
- Applied statistical methods (hypothesis testing and significance levels, regression, bootstrap and sampling methods) to the ensemble, quantifying the significance and robustness of results against internal variability (noise) across forcing scenarios
- Modeled nonlinear, stochastic dynamics and regime transitions across 6 CO2 forcing scenarios, identifying a critical tipping threshold through large-ensemble numerical simulation

Software Engineering Intern, Formosa Climate Smart Service, *Taipei, Taiwan* — July 2023 – Sept 2023

- Developed a web platform (Python, HTML/JavaScript) implementing Taiwan's building energy-efficiency rating, computing Energy Use Intensity scoring scales and efficiency grades with utility-data reliability checks
- Extracted floor-zone areas from DWG architectural plans with AutoCAD scripting and prototyped 3D ArcGIS maps of building EUI

Visiting Student Intern, Lamont-Doherty Earth Observatory, *Palisades, NY, USA* — June 2022 – Aug 2022

- Applied variance decomposition in Python to roughly 100 ensemble members from 6 climate models, partitioning uncertainty in projected Arctic change into internal variability and inter-model spread across seasons.
- Applied bootstrap resampling and significance testing to large-ensemble simulations to quantify the robustness of a physical signal against internal variability (noise)

Publications

1. Previdi, M., **Wu, Y.-T.**, Polvani, L.M., England, M.R., Sigmond, M., Liang, Y.-C. Substantial Twentieth-Century Global Sea Level Rise Caused by Ozone-Depleting Substances. Submitted to Journal of Climate (under revision).
2. **Wu, Y.-T.**, Previdi, M., Romanou, A., Lerner, P., Polvani, L.M. (2026). Arctic Climate Tipping Points under Delayed CO2 Emissions Mitigation. Communications Earth and Environment.
3. Liang, Y.-C., Polvani, L.M., Previdi, M., Dong, Y., **Wu, Y.-T.**, England, M.R., Griffith, S.M. (2026). The Importance of Non-CO2 Greenhouse Gases to Arctic Warming and Sea-Ice Loss. Geophysical Research Letters, 53, e2026GL122867.
4. **Wu, Y.-T.**, Liang, Y.-C., Previdi, M., Polvani, L.M., England, M.R., Sigmond, M., Lo, M.-H. (2024). Stronger Arctic Amplification from Anthropogenic Aerosols than from Greenhouse Gases. npj Climate and Atmospheric Science, 7(1), 142.
5. **Wu, Y.-T.**, Liang, Y.-C., Kuo, Y.-N., Lehner, F., Polvani, L.M., Previdi, M., Lo, M.-H., Lan, C.-W. (2023). Exploiting SMILEs and the CMIP5 Archive to Understand Arctic Climate Change Seasonality and Uncertainty. Geophysical Research Letters, 50(2), e2022GL100745.

Awards

- Government Scholarship to Study Abroad, Ministry of Education, Taiwan (June 2026)
- Nominated for membership, Sigma Xi, The Scientific Research Honor Society, USA (January 2026)
- Excellent Undergraduate Research Project Award, Ministry of Science and Technology, Taiwan (June 2023)
- Undergraduate Research Project Grant, Ministry of Science and Technology, Taiwan (June 2022)

Professional Activities

- Journal reviewer: Nature Communications (2026)
- 12 scientific presentations at international conferences (2021–2025)
- 12 teaching assistantships, including Partial Differential Equations, Numerical Analysis, and Applied Mathematics